

Team Name: Elite Coders

Team Members:

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Project Name: Jet Cultivation

Experiment 6: ER Diagram

AIM:

Creating ER Diagram in draw.io and Uploading to Azure DevOps

PROCEDURE:

Step 1: Open draw.io (diagrams.net)

1. Open browser and go to: <https://app.diagrams.net>
2. Choose Device / Google Drive
3. Click Create New Diagram

Step 2: Create New Diagram

1. Enter file name (e.g., *ER Diagram*)
2. Select Blank Diagram
3. Click Create

Step 3: Add Entities (Rectangles)

1. Use Rectangle shape
2. Add entities:
 - Crops
 - Soil
 - Farmer
 - Optimal Centers
 - Crop Journey
 - River
 - Irrigation Method

Step 4: Add Attributes (Ovals)

1. Use Ellipse shape
2. Connect attributes to respective entities

Crops

- crop_id

- crop_name
- season
- water_req

Crop Journey

- journey_id
- stage_name
- description
- duration_days

Soil

- soil_id
- soil_type
- pH_level
- nutrients

Farmer

- farmer_id
- name
- phone
- email
- password
- location

Optimal Centers

- center_id
- center_name
- location
- address
- phone
- services

River

- river_id
- river_name
- status
- water_level
- water_flow

Irrigation Method

- irrigation_id
- irrigation_name
- description

Step 5: Add Relationships (Diamonds)

1. Use Diamond shape
2. Add relationships:
 - Crops → Has → Crop Journey
 - Crops → Supports → Soil
 - Soil → Has → Farmer
 - Farmer → Views → Optimal Centers
 - River → Determines → Irrigation Method

Step 6: Add Connections

1. Use straight lines
2. Connect:
 - Entity ↔ Relationship
 - Entity ↔ Attributes

Step 7: Add Cardinality

1. Mention:
 - 1 (one)
 - - (many)
2. Example:
 - One soil → many crops
 - One farmer → many centers

Step 8: Arrange Diagram

1. Keep Crops at top
2. Soil below crops
3. Farmer below soil
4. Other entities around neatly
5. Maintain spacing

Step 9: Save Diagram

1. Click File → Save As (.drawio)
2. Export as:
 - PNG / JPG

Step 10: Open Azure DevOps

1. Go to your project
2. Navigate to:
3. Wiki or Repos

Step 11: Upload Diagram

Wiki:

1. Open page
2. Click Edit
3. Insert Image
4. Upload diagram

Repos:

1. Click Upload Files
2. Add image
3. Commit

Step 12: Verify

1. Open uploaded diagram
2. Check clarity and correctness

Result

The ER Diagram for Jet Cultivation System is successfully created using draw.io and uploaded to Azure DevOps.

